Layer2Meetup@Tokyo Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Wallet Governance

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The Global Network for Blockchain Stakeholders™

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Layer2 Meetup@Tokyo

Keynote Session Report: Wallet Governance

Date: Friday, September 20th, 2024

Location: Tokyo

Duration: Approx. 90 minutes

Summary

Presentation of Wallet Governance and Policy Study Report by BGIN

1. Convergence of Wallet Use Cases:

- **Question**: One of the key questions raised was whether financial and non-financial use cases for wallets should come together in a single wallet or remain separate.
- **Discussion**: Several participants pointed out that from a user experience perspective, separating financial and non-financial functions would be detrimental, as it would complicate adoption. This sparked a discussion about whether identity and financial transactions would converge naturally, as trust and reputation systems evolve.

2. AI Integration in Wallets:

- **Question**: A question was raised about whether AI agents will become primary users of blockchain wallets in the future and what the implications might be for **AML/CFT** compliance.
- **Discussion**: Participants considered how AI agents might impact financial transactions and whether AI could be held accountable for its actions. The discussion explored how AI-driven transactions could complicate KYC/AML processes, and whether there would be a need for signatures proving that a human authorized an AI transaction.

3. Governance Wallets and Voting:

- **Question**: The discussion touched on whether wallets should be used for governance activities, such as voting, and what security considerations should be taken into account.
- **Discussion**: Some participants argued that governance wallets might sit at the intersection of finance and identity, but there was debate about whether wallets should have a specific governance function or if existing wallets (used for finance and identity) could be adapted for governance. Concerns about anonymity and security in voting were also raised, especially in governance systems where cryptographic signatures are essential.

4. On-Chain vs. Off-Chain Wallet Governance:

- **Question**: Should wallet governance take place on-chain, or is there a need for off-chain solutions to manage governance use cases more effectively?

- **Discussion**: The debate focused on the pros and cons of on-chain governance, which offers transparency but comes with challenges such as vulnerability to attacks. Off-chain governance, on the other hand, might offer more flexibility but could sacrifice transparency. Participants discussed the potential for using zero-knowledge proofs and decentralized storage solutions to protect data while maintaining the security of wallets used for governance.

5. Identity and Privacy in Wallets:

- **Question**: Should wallets be focused solely on identity, or should identity be a starting point for integrating other use cases (such as finance and governance)?
- **Discussion**: This sparked a conversation about whether identity and reputation drive financial transactions, or if finance leads to the creation of reputation and identity. Suggestions included using wallets to store and manage identity information securely, with features like granular consent and selective disclosure to protect privacy. There was agreement that privacy-enhancing technologies, like zero-knowledge proofs, could be crucial in the future of identity wallets.

Presentation about Biometric Key Management

- **Question**: One of the participants asked if splitting a transaction into two stages (ordering and authorizing with a fingerprint) could be considered similar to a biometric-bound wallet.
- **Answer**: The presenter clarified that biometric-locked wallets activate a stored key with biometrics but do not directly generate signatures from the biometric data, as biometric-bound wallets do.

Presentation about Accountable Wallet

1. **KYC Responsibility – Sender or Receiver?**
    - **Question**: A participant asked about who holds the responsibility for KYC (Know Your Customer) compliance—whether it's the sender, the receiver, or the party initiating the transaction.
    - **Response from VASP Chief Compliance Officer**: The VASP (Virtual Asset Service Provider) representative clarified that KYC responsibilities typically lie with the VASP, not individual receivers. If funds from an illegal source are detected, VASPs like Coinbase might reject the reception of funds. Individual receivers can also take steps to mitigate risk by checking the compliance of the sender with AML/CFT measures.
2. **Reputation Systems for No-Custody Payment Systems:**
    - **Question**: A participant asked whether there has been any discussion around reputation systems for decentralized, no-custody payment systems. Specifically, they were curious about creating a reputation-based revocation system without a central authority.
    - **Discussion**: Bitcoin Core developers tend to avoid reputation systems, preferring privacy-focused solutions like **Silent Payments** (a feature not yet in Bitcoin Core), which make wallet addresses untraceable. The speaker emphasized that while a reputation

system could help filter out bad actors, it would clash with the privacy ethos of many in the Bitcoin community.

3. **Fungibility and the Future of Bitcoin:**
    - **Bitcoin Becoming Non-Fungible?**: One participant raised an interesting point—if reputation systems and accountability are introduced, Bitcoin could become like non-fungible tokens (NFTs), where "clean" Bitcoins are valued higher than "dirty" Bitcoins. This could complicate the user experience, with people needing to check the legitimacy of the coins they receive.
    - **Developer Perspective**: A participant explained that many Bitcoin Core developers are wary of creating a system where one Bitcoin is worth less than another based on its history. They want to preserve fungibility, where all Bitcoins are treated equally, and are focused on developing technologies that improve privacy and fungibility.
    - **Potential for a Divided Ecosystem**: If accountability and compliance become integral to cryptocurrency ecosystems, a new DeFi space could emerge where "clean" Bitcoins are valued higher than "dirty" ones. This would create a market where fungibility is compromised, raising concerns about the long-term viability of such a system.
4. **Balancing Accountability and Privacy:**
    - **Challenges with Balancing Both**: The key challenge highlighted in the discussion was balancing the desire for privacy (e.g., through anonymous transactions) with the need for accountability to meet regulatory standards. The introduction of accountability mechanisms could divide the crypto economy into those who comply and those who do not.
    - **Onion Routing in the Lightning Network**: The Lightning Network was mentioned as an example where privacy technologies like **onion routing** allow transactions to remain anonymous. Yet, some exchanges, like Gemini, still support privacy coins (e.g., Zcash), while others, like Coinbase, choose not to deal with them due to regulatory complexities.
5. **Impact on Bitcoin Core Development:**
    - **Resistance to Change**: A participant expressed concerns that Bitcoin Core developers might lose relevance if they continue to resist changes around accountability. As more people move to using exchanges or third-party services, privacy and anonymity may become less prioritized.
    - **Regulatory and Market Trends**: Privacy coins like Zcash and Monero have relatively low usage, indicating that most users may prioritize convenience over privacy, favoring compliance-driven solutions. This suggests that developers focused solely on privacy may be missing a broader market demand.
6. **Future of No-Custody Systems and Exchanges:**
    - **VASP Challenges**: Representatives from VASPs shared their struggles in accepting funds from self-hosted wallets. Each VASP uses different metrics and blockchain analysis tools, which complicates the implementation of uniform due diligence standards.

- **Proposed Solutions**: Some participants felt that reconciling due diligence policies across VASPs could help improve the situation. However, if regulatory trends continue, it's possible that VASPs may increasingly reject self-hosted wallets.